Nermine Moumen

Industrial Chemical Engineering Student

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EDUCATION

National Institute of Applied Sciences and Technology (INSAT) *2022–Present*

- Final-year Industrial Chemical Engineering student, specialized in Process and Environment, with a broad foundation spanning Thermodynamics, Fluid Mechanics, Energy & Mass Balances, Renewable energy and Materials Science. Seeking a final-year internship focused on process simulation and techno-economic evaluation of sustainable energy conversion routes.

PROFESSIONAL EXPERIENCE

SEGULA Technologies – Electrochemical Battery Modeling Internship *06/2026–08/2026*

- Modeled a **Li-ion cell** (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Solved the governing PDEs using the **Finite Volume Method** (FVM) with Crank–Nicolson time integration.
- Optimized the mesh and diffusion coefficient values, reducing the RMSE error **from 202.13 mV to 40.5 mV**.
- Extracted and processed **experimental discharge data** from a database to validate MATLAB and COMSOL results.

Control Union – Quality Control Internship *08/2025–09/2025*

- Analyzed lubricants for viscosity and heavy metals **(ICP-OES)**.
- Prepared and interpreted **analytical reports** to verify compliance with ASTM D5185 and **ISO 17025** standards.

ETAP – Water Treatment and Petrochemistry Internship *07/2024–08/2024*

- Studied water treatment methods **(coagulation-flocculation, ion exchange)** for contaminant and corrosion control.
- Performed organic geochemistry techniques **(GC, GC-MS, Rock-Eval pyrolysis)** for hydrocarbon **characterization**.

ADDITIONAL EXPERIENCE

Cascade – Online Chemistry Teacher (Part-time) *2024–Present*

- Provide individualized online chemistry tutoring for French Terminale students, covering exam-level concepts.

ACADEMIC PROJECTS

Techno-Environmental Analysis of Ethylene Production: Steam Cracking vs. Bioethanol Dehydration

- Simulated both routes in Aspen Plus V14, comparing reactor design, separation train, and energy integration, achieving a purified ethylene stream at **99.95 mol% purity**.
- Conducted a cradle-to-gate **Life Cycle Assessment** in **OpenLCA** using Aspen Plus outputs as inventory data.

Prediction of Liquid–Liquid Equilibria and Cross-flow and Countercurrent Extraction for Ternary Systems

- Formulated the LLE thermodynamic system and solved it via Rachford-Rice and **arc-length continuation**.
- Reproduced 7 Mathematica notebooks proving equivalence of the **Hunter-Nash and McCabe-Thiele** methods.
- Reimplemented workflow in **Python**, MATLAB, and Mathematica (spline/fsolve, SciPy/mpltern, Interpolation/FindRoot).

Ethylene Oxide Production Process Simulation using Aspen HYSYS

- Simulated the ethylene oxide production process using NRTL-RK thermodynamics and **Langmuir-Hinshelwood kinetics**.
- Optimized operating conditions via sensitivity analysis, achieving 96% recovery and **97.14% final purity.**

Hydrogen Energy Cycle Modeling: PEM Fuel Cells and Electrolysis

- Simulated isothermal and nonisothermal PEMFC behavior in COMSOL Multiphysics, achieving a cell voltage of 0.584 V.
- Modeled a **3D PEM electrolyzer** in Ansys Fluent, achieving 1.40 L/h H2 production and identifying the oxygen evolution reaction as the dominant kinetic bottleneck.

TECHNICAL SKILLS

- **Simulation:** Aspen Plus, Aspen HYSYS, COMSOL Multiphysics
- **Programming:** $MATLAB, Mathematica, Python$
- **Data & Office:** Pack Microsoft Office, OpenLCA
- **Standards:** $ISO9001, ISO14001, ISO45001, ISO17025, ISO14040$
- **Lab Techniques:** Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM), Electrochemistry (Cyclic voltammetry, Polarography).

EXTRACURRICULAR ACTIVITIES

- **IEEE INSAT** – Chair of EMBS; Organization team leader of TRSYP 3.0; Social activities manger of MENASYP
- **THEATRO INSAT** – Member of the dancing team

LANGUAGES

English (Advanced) French (Advanced) Arabic (Native)

CERTIFICATES

Aspen HYSYS – MATLAB